

Lab 2 : Docker Container Lifecycle and Management Using Nginx

Step 1: Check Docker Installation

docker --version

Output:

```
C:\>docker --version
Docker version 29.3.1, build c2be9cc
```

Step 2: Search for Nginx

docker search nginx

Output:

```
C:\>docker search nginx
NAME                                DESCRIPTION                                STARS    OFFICIAL
nginx                               Official build of Nginx.                  21337    [OK]
nginx/nginx-ingress                 NGINX and NGINX Plus Ingress Controllers fo... 121
nginx/nginx-prometheus-exporter     NGINX Prometheus Exporter for NGINX and NGIN... 52
nginx/nginx-ingress-operator        NGINX Ingress Operator for NGINX and NGINX P... 4
nginx/nginxaas-loadbalancer-kubern... 1
bitnamicharts/nginx                Bitnami Helm chart for NGINX Open Source      4
ubuntu/nginx                       Nginx, a high-performance reverse proxy & we... 141
kasmweb/nginx                       An Nginx image based off nginx:alpine and in... 9
rancher/nginx                       4
linuxserver/nginx                  An Nginx container, brought to you by LinuxS... 236
dtagdevsec/nginx                   T-Pot Nginx                                0
paketobuildpacks/nginx             0
vmware/nginx                        3
gluu/federation/nginx              A customized NGINX image containing a consu... 1
cleanstart/nginx                   Secure by Design, Built for Speed, Hardened ... 0
antrea/nginx                       Nginx server used for Antrea e2e testing      0
activestate/nginx                  ActiveState's customizable, low-to-no vulner... 0
intel/nginx                         0
docksal/nginx                      Nginx service image for Docksal              1
geokrety/nginx                     Our customized nginx image                   0
circleci/nginx                     This image is for internal use                2
ilios/nginx                        Nginx customized to run Ilios along with the... 0
corpusops/nginx                    https://github.com/corpusops/docker-images/  1
wayofdev/nginx                     Nginx Docker image for PHP development. SSL-... 0
dockette/nginx                     Nginx SSL / HSTS / HTTP2                    3
```

Step 3: Download Nginx

docker pull nginx

Output:

```
C:\>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
062e450697fa: Pull complete
39694cde734e: Pull complete
3b9f3509a826: Pull complete
0b19be717994: Pull complete
8d2092eabbc5: Pull complete
6a24a6ebcf30: Pull complete
45ef33eb91c8: Pull complete
Digest: sha256:c90e4ea5905ee69515d84cd5520e11d204df06cce71a312666cebbbc4b9b68267
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

Step 4: View Images

docker images

Output:

```
C:\>docker images
```

ID	DISK USAGE	CONTENT SIZE	EXTRA	
calculator-app:latest	931b6c297e28	1.12GB	0B	
hello-docker:latest	5a7cbf5cfdfe	141MB	0B	
nginx:latest	8870c81b2834	161MB	0B	
simple-docker-app:latest	bee3035e96ea	134MB	0B	

 Info →  In UseIMAGE

Step 5: Run Nginx

docker run nginx

Output:

```

C:\>docker run nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 09:18:41 [notice] 1#1: using the "epoll" event method
2026/07/14 09:18:41 [notice] 1#1: nginx/1.31.2
2026/07/14 09:18:41 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 09:18:41 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/07/14 09:18:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 09:18:41 [notice] 1#1: start worker processes
2026/07/14 09:18:41 [notice] 1#1: start worker process 29
2026/07/14 09:18:41 [notice] 1#1: start worker process 30
2026/07/14 09:18:41 [notice] 1#1: start worker process 31
2026/07/14 09:18:41 [notice] 1#1: start worker process 32
2026/07/14 09:18:41 [notice] 1#1: start worker process 33
2026/07/14 09:18:41 [notice] 1#1: start worker process 34
2026/07/14 09:18:41 [notice] 1#1: start worker process 35
2026/07/14 09:18:41 [notice] 1#1: start worker process 36
2026/07/14 09:18:41 [notice] 1#1: start worker process 37
2026/07/14 09:18:41 [notice] 1#1: start worker process 38
2026/07/14 09:18:41 [notice] 1#1: start worker process 39
2026/07/14 09:18:41 [notice] 1#1: start worker process 40

```

Step A

A new version of Docker Desktop is available. Your current version is 110 days old.

Search

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Containers

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1.58 GB / 1.42 GB in use 6 images

Last refresh: 35 minutes ago

Search

	Name	Tag	Image ID	Created	Size	Actions
☐	calculator-app	latest	931b6c297e28	10 days ago	1.12 GB	
☐	<none>	<none>	cf587c96b09	10 days ago	1.12 GB	
☐	<none>	<none>	5881574db5b0	10 days ago	1.12 GB	
☐	simple-docker-app	latest	bee3035e96ea	10 days ago	133.8 MB	
☐	Copy to clipboard	latest	5a7cbf5cdfde	3 months ago	140.86 MB	
☐	nginx	latest	8870c81b52834	8 hours ago	161.3 MB	

Showing 6 items

Engine running

RAM 0.96 GB CPU 0.00% Disk: 3.12 GB used (limit 1006.85 GB)

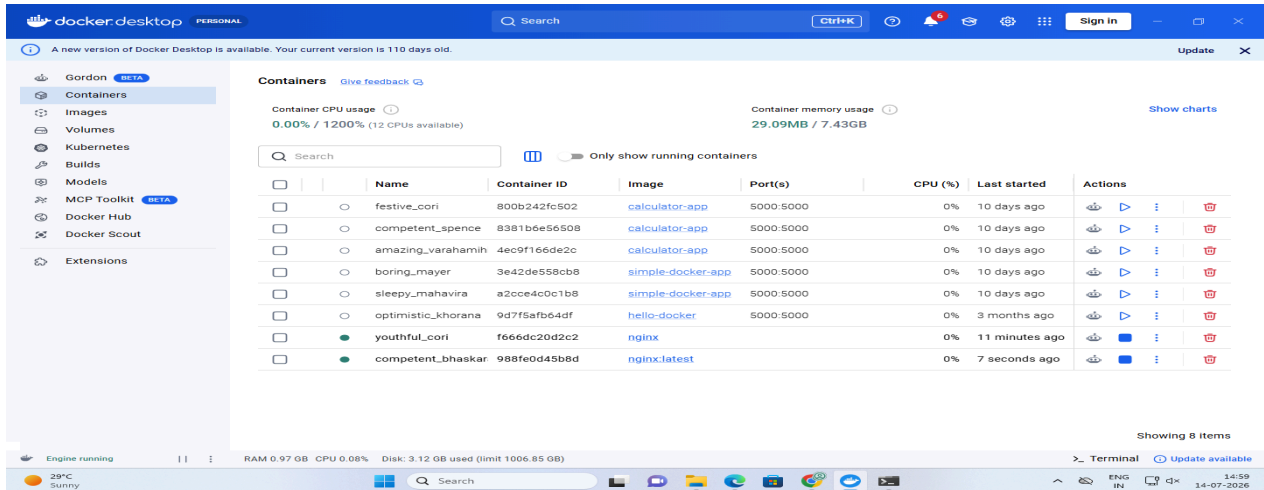
Terminal Update available

29°C Sunny

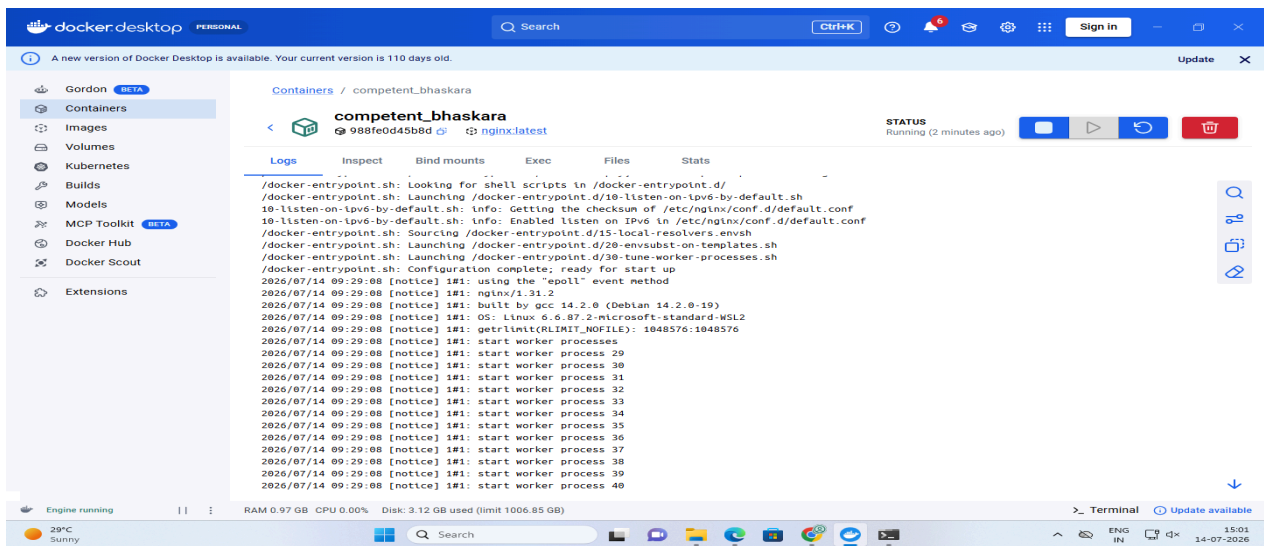
Search

ENG IN 14:57 14-07-2026

Step B



Step C



Step D

```
Command Prompt
2026/07/14 09:18:41 [notice] 1#1: start worker process 39
2026/07/14 09:32:14 [notice] 1#1: start worker process 40
2026/07/14 09:32:14 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/07/14 09:32:14 [notice] 29829: exiting
2026/07/14 09:32:14 [notice] 30830: exiting
2026/07/14 09:32:14 [notice] 33833: exiting
2026/07/14 09:32:14 [notice] 34834: exiting
2026/07/14 09:32:14 [notice] 35835: exiting
2026/07/14 09:32:14 [notice] 36836: exiting
2026/07/14 09:32:14 [notice] 38838: exit
2026/07/14 09:32:14 [notice] 33831: exit
2026/07/14 09:32:14 [notice] 31831: exiting
2026/07/14 09:32:14 [notice] 34834: exit
2026/07/14 09:32:14 [notice] 35835: exit
2026/07/14 09:32:14 [notice] 36836: exit
2026/07/14 09:32:14 [notice] 31831: exit
2026/07/14 09:32:14 [notice] 32832: exiting
2026/07/14 09:32:14 [notice] 32832: exit
2026/07/14 09:32:14 [notice] 39839: exiting
2026/07/14 09:32:14 [notice] 39839: exit
2026/07/14 09:32:14 [notice] 38838: exiting
2026/07/14 09:32:14 [notice] 37837: exiting
2026/07/14 09:32:14 [notice] 40840: exiting
2026/07/14 09:32:14 [notice] 38838: exit
2026/07/14 09:32:14 [notice] 40840: exit
2026/07/14 09:32:14 [notice] 37837: exit
2026/07/14 09:32:14 [notice] 1#1: signal 14 (SIGALRM) received
2026/07/14 09:32:14 [notice] 1#1: signal 17 (SIGCHLD) received from 31
2026/07/14 09:32:14 [notice] 1#1: worker process 31 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: worker process 37 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 09:32:14 [notice] 1#1: signal 17 (SIGCHLD) received from 32
2026/07/14 09:32:14 [notice] 1#1: worker process 32 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: worker process 33 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: worker process 40 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 09:32:14 [notice] 1#1: signal 17 (SIGCHLD) received from 39
2026/07/14 09:32:14 [notice] 1#1: worker process 35 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: worker process 36 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: worker process 39 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 09:32:14 [notice] 1#1: signal 17 (SIGCHLD) received from 35
2026/07/14 09:32:14 [notice] 1#1: signal 17 (SIGCHLD) received from 30
2026/07/14 09:32:14 [notice] 1#1: worker process 29 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: worker process 30 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: worker process 34 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: worker process 38 exited with code 0
2026/07/14 09:32:14 [notice] 1#1: exit

C:\>
```

Step 6: Detached Mode

docker run -d nginx

Output:

```
C:\>docker run -d nginx
3198705cd76d32095e93170ca2cc8b41dc6ca4b37115d4fdb3e7a9539551eb55
```

Step 7: Name the Container

docker run -d --name mynginx nginx

Output:

```
C:\>docker run -d --name mynginx nginx
b3be5333f0521b68aefc19a38f21d6752007c5894da769d93831698a33c13326
```

Step 8: Publish a Port

```
docker run -d --name webserver -p 8080:80 nginx
```

Output:

```
C:\>docker run -d --name webserver -p 8080:80 nginx
42d30f40cdebac080dc78bf68363eb2d151585f8d0b9df1ae62b0914eae3ecbb
```

Step 9: List Running Containers

```
docker ps
```

Output:

```
C:\>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
42d30f40cdeb   nginx    "/docker-entrypoint..." About a minute ago Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   webserver
b3be5333f052   nginx    "/docker-entrypoint..." 4 minutes ago Up 4 minutes       80/tcp                               mynginx
3198705cd76d   nginx    "/docker-entrypoint..." 6 minutes ago Up 6 minutes       80/tcp                               cranky_hodgkin
988fe0d45b8d   nginx:latest "/docker-entrypoint..." 13 minutes ago Up 13 minutes       80/tcp                               competent_bhaskara
```

Step 10: Stop Container

```
docker stop mynginx
```

Output:

```
C:\>docker stop mynginx
mynginx
```

Step 11: Start Container Again

docker start mynginx

Output:

```
C:\>docker start mynginx
mynginx
```

Step 12: Restart

docker restart mynginx

Output:

```
C:\>docker restart mynginx
mynginx
```

Step 13: View Logs

docker logs mynginx

Output:

```
C:\>docker logs mynginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 09:37:33 [notice] 1#1: using the "epoll" event method
2026/07/14 09:37:33 [notice] 1#1: nginx/1.31.2
2026/07/14 09:37:33 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 09:37:33 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/07/14 09:37:33 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 09:37:33 [notice] 1#1: start worker processes
2026/07/14 09:37:33 [notice] 1#1: start worker process 29
2026/07/14 09:37:33 [notice] 1#1: start worker process 30
```

Step 14: Enter the Container

```
docker exec -it mynginx bash
```

Output:

```
C:\>docker exec -it mynginx bash
root@b3be5333f052:/# pwd
/
root@b3be5333f052:/# ls
bin      dev          docker-entrypoint.sh  home  lib64  mnt  proc  run  srv  tmp  var
boot    docker-entrypoint.d  etc                lib   media  opt  root  sbin  sys  usr
root@b3be5333f052:/# |
```

Step 15: Remove the Container

```
docker rm mynginx
```

Output:

```
C:\>docker rm mynginx
Error response from daemon: cannot remove container "mynginx": container is running: stop the container before removing or force remove
```

Step 16: Remove the Image

```
docker rmi nginx
```

Output:

```
C:\>docker rmi nginx
Error response from daemon: conflict: unable to remove repository reference "nginx" (must force) - container f666dc20d2c2 is using its referenced image 8870c81b2834
```